MakeNoise<VectorDomain<AtomDomain<T>>, LpDistance<P, QI>, MO> for IntExpFamily<P>

Michael Shoemate (@Shoeboxam)

This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of the implementation of MakeNoise over vectors for IntExpFamily in mod.rs at commit 286ce37ce (outdated[1]).

This mechanism samples from the IntExpFamily distribution, where the tails are clipped to the smallest and largest representable integers in the native integer type. This is done by first sampling from the ZExpFamily, the equivalent distribution supported on all integers, and then clipping the tails by clamping. The clamping is done by saturating the cast of the sampled value to the native integer type.

1 Hoare Triple

Precondition

Compiler-verified

- Generic T implements trait Integer and SaturatingCast<IBig> The saturating cast is for infallible postprocessing of big ints back to type T.
- Const-generic P is of type usize
- Generic QI implements trait Integer
- Generic MO implements trait Measure
- Type IBig implements trait From<T>. This infallible exact cast is for converting integers to big ints in the preprocessing transformation.
- Type RBig implements trait TryFrom<QI>. This is for fallible casting from input sensitivity of type QI to a rational in the privacy map.
- Type ZExpFamily<P> implements trait NoisePrivacyMap<LpDistance<P, RBig>, MO>. This bound requires that it must be possible to construct a privacy map for this combination of noise distribution, distance type and privacy measure.

Caller-verified

None

[1] See new changes with `git diff 286ce37ce..58d6f416 rust/src/measurements/noise/nature/integer/mod.rs`

Pseudocode

```
class IntExpFamily:
    def make_noise(
        self, input_space: tuple[VectorDomain[AtomDomain[T]], LpDistance[P, QI]]
    ) -> Measurement[VectorDomain[AtomDomain[T]], T, LpDistance[P, QI], MO]:
        distribution = ZExpFamily(
            scale=integerize_scale(self.scale, 0)
        )  #

        t_int = make_int_to_bigint(input_space)
        m_noise = distribution.make_noise(t_int.output_space())
        return t_int >> m_noise >> then_saturating_cast()
```

Postcondition

Theorem 1.1. For every setting of the input parameters (`self, input_space, MO, T, P, QI`) to `make_noise` such that the given preconditions hold, `make_noise` raises an error (at compile time or run time) or returns a valid measurement. A valid measurement has the following properties:

1. (Data-independent runtime errors). For every pair of members x and x' in `input_domain`, `invoke`(x) and `invoke`(x') either both return the same error or neither return an error.

2. (Privacy guarantee). For every pair of members x and x' in `input_domain` and for every pair (`d_in`, `d_out`), where `d_in` has the associated type for `input_metric` and `d_out` has the associated type for
`output_measure`, if x, x' are `d_in`-close under `input_metric`, `privacy_map`(`d_in`) does not raise an error, and `privacy_map`(`d_in`) $=$ `d_out`, then `function`(x), `function`(x') are `d_out`-close under `output_measure`.

Proof. Line 7 constructs a new random variable following a distribution equivalent to `IntExpFamily`, but without clipped tails.

Neither constructor `make_int_to_bigint` nor `MakeNoise.make_noise` have manual preconditions, and the postconditions guarantee a valid transformation and valid measurement, respectively. `then_saturating_cast` also does not have preconditions, and its postcondition guarantees that it returns a valid postprocessor.

The chain of a valid transformation, valid measurement and valid postprocessor is a valid measurement. □